THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · NCERT-TYPE UNIT TEST SET 3

Chapters 9–13: **Cell: The Building Block of Life, Tissues in Action, Patterns in Life: Diversity and Classification, Reproduction: How Life Continues, Earth as a System: Energy, Matter and Life** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Virchow added to the cell theory that: 1

(a) cells are made of atoms **(b)** cells are visible to the eye
(c) new cells arise from pre-existing cells **(d)** all cells have walls

2. The epithelium lining the blood vessels and alveoli is: 1

(a) squamous epithelium **(b)** cuboidal epithelium
(c) ciliated epithelium **(d)** columnar epithelium

3. Hydra and jellyfish belong to: 1

(a) Porifera **(b)** Annelida
(c) Cnidaria (Coelenterata) **(d)** Arthropoda

4. Fusion of the male and female gametes is called: 1

(a) pollination **(b)** germination **(c)** fertilisation **(d)** budding

5. The ozone layer protects us from: 1

(a) infrared radiation **(b)** visible light
(c) harmful ultraviolet radiation **(d)** sound

6. The organelle called the 'powerhouse of the cell' is the: 1

(a) Golgi apparatus **(b)** lysosome **(c)** mitochondrion **(d)** chloroplast

7. Muscles in the walls of the intestine are: 1

(a) cardiac muscles **(b)** striated muscles
(c) voluntary muscles **(d)** smooth (involuntary) muscles

8. Frogs and toads belong to: 1

(a) Amphibia **(b)** Pisces **(c)** Aves **(d)** Reptilia

9. The ova are produced in the: 1

(a) ovaries **(b)** testes **(c)** uterus **(d)** vagina

10. A biogeochemical cycle is the: 1

(a) cycle of the Moon
(b) cycle of seasons
(c) cycle of day and night
(d) cycling of nutrients through living and non-living parts of the Earth

11. Chromosomes are made of: 1

(a) only fat **(b)** DNA and proteins **(c)** only water **(d)** only starch

12. Grass regrows after being cut because of the: 1

(a) apical meristem **(b)** lateral meristem

(c) intercalary meristem **(d)** cork cambium

13. The binomial system of nomenclature was introduced by: 1

(a) Aristotle **(b)** Darwin **(c)** Carolus Linnaeus **(d)** Mendel

14. Yeast reproduces mainly by: 1

(a) fragmentation **(b)** binary fission **(c)** budding **(d)** spores only

15. All water on the Earth, in oceans, rivers, ice and groundwater, forms the: 1

(a) biosphere **(b)** geosphere **(c)** atmosphere **(d)** hydrosphere

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. What is plasmolysis? 2

17. State the functions of the epithelial tissue. 2

18. Distinguish platyhelminthes from nematodes. 2

19. Why do animals with external fertilisation lay more eggs? 2

20. State two roles of the atmosphere in supporting life. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Explain the importance of the nitrogen-fixing bacteria in agriculture. 3

22. Describe an activity to show osmosis using a potato. 3

23. Describe the structure and functions of the epidermis in a leaf and in a root. 3

24. Why are fungi called decomposers? 3

OR

Describe budding in Hydra and yeast. 3

25. Describe the natural greenhouse effect and the enhanced greenhouse effect. 3

OR

What is meant by a hypotonic, hypertonic and isotonic solution? Give the effect on a cell. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Compare asexual and sexual reproduction in five points with examples. 5

OR

Describe the nitrogen cycle with the processes: fixation, absorption, assimilation, decay, and denitrification. 5

27. Describe the structure of a typical plant cell and state the three structures that distinguish it from an animal cell. 5

OR

Describe the simple permanent tissues with structure and function: parenchyma, collenchyma and sclerenchyma. 5

END OF QUESTION PAPER **Total: 50 Marks**

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